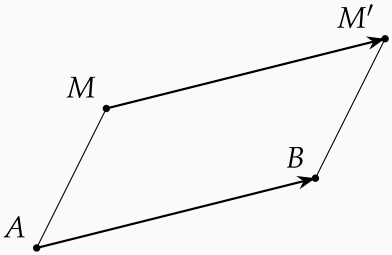
Vectors in the plane

Lecture and worked examples

Translation and vectors

The translation that transforms A into B associates with every point M the unique point M' such that $ABM'M$ is a parallelogram, possibly flat. It is called the translation by the vector $\overrightarrow{AB}$. A nonzero vector has a direction, a sense and a norm. $\|\overrightarrow{AB}\| = AB$. The vector $\overrightarrow{AA}$ is the zero vector, written $\vec{0}$; it has neither direction nor sense.



Translation and vectors

$\overrightarrow{AB} = \overrightarrow{CD}$ when the translation by the vector $\overrightarrow{AB}$ transforms C into D. Two nonzero vectors are equal if and only if they have the same direction, the same sense and the same norm. The opposite vector of $\overrightarrow{AB}$ is $\overrightarrow{BA}$: $\overrightarrow{BA} = -\overrightarrow{AB}$.

Translation and vectors

Draw a segment $[AB]$ and place a point C outside (AB). Construct D such that $\overrightarrow{AB} = \overrightarrow{CD}$. Name the parallelogram obtained. Construct E such that $\overrightarrow{CE} = -\overrightarrow{AB}$.

Worked answer: Translation and vectors

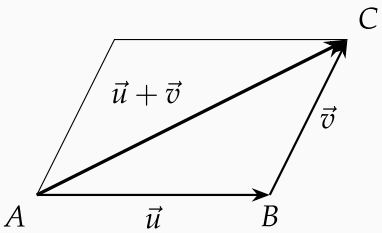
D is constructed by carrying over from C the direction, sense and length of $\overrightarrow{AB}$. The parallelogram is $ABDC$, in that order. E is the reflection of D in the point C, because $\overrightarrow{CE} = -\overrightarrow{CD}$.

Sum, difference and Chasles's relation

The sum $\vec{u} + \vec{v}$ corresponds to carrying out the translations by the vectors $\vec{u}$ and $\vec{v}$ one after the other. For all points A, B, C:

$$\boxed{\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}} \qquad \text{(Chasles's relation)}.$$

We have $\vec{u} - \vec{v} = \vec{u} + (-\vec{v})$ and $\overrightarrow{AB} + \overrightarrow{BA} = \vec{0}$.



Sum, difference and Chasles's relation

$ABDC$ is a parallelogram if and only if $\overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{AC}$.

Sum, difference and Chasles's relation

Simplify: $\overrightarrow{AB} + \overrightarrow{BC}$; $\overrightarrow{AB} + \overrightarrow{CA}$; $\overrightarrow{AB} + \overrightarrow{BC} + \overrightarrow{CD}$; $\overrightarrow{AB} - \overrightarrow{AC}$.

Worked answer: Sum, difference and Chasles's relation

$\overrightarrow{AC}$; $\overrightarrow{CA} + \overrightarrow{AB} = \overrightarrow{CB}$; $\overrightarrow{AD}$; $\overrightarrow{AB} + \overrightarrow{CA} = \overrightarrow{CB}$. The commutativity of addition allows us to reorder the vectors.

Sum, difference and Chasles's relation

In a parallelogram $ABCD$, justify that $\overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AC}$.

Worked answer: Sum, difference and Chasles's relation

In $ABCD$, $\overrightarrow{AD} = \overrightarrow{BC}$. So $\overrightarrow{AB} + \overrightarrow{AD} = \overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$.

Product of a vector by a real number

For $k \in \mathbb{R}$, $k\vec{u} = \vec{0}$ if $k = 0$ or $\vec{u} = \vec{0}$. Otherwise, $k\vec{u}$ has the same direction as $\vec{u}$, the same sense if $k > 0$, the opposite sense if $k < 0$, and its norm is $|k|\|\vec{u}\|$.

$$(k+\ell)\vec{u} = k\vec{u} + \ell\vec{u}, \quad k(\vec{u}+\vec{v}) = k\vec{u} + k\vec{v}, \quad k(\ell\vec{u}) = (k\ell)\vec{u}.$$

Product of a vector by a real number

For a vector $\vec{u}$ of norm 4 cm, represent $2\vec{u}$, $-\vec{u}$ and $\frac{1}{2}\vec{u}$ from the same origin. State their norms.

Worked answer: Product of a vector by a real number

The norms are 8 cm, 4 cm, 2 cm. The vectors $2\vec{u}$ and $\frac{1}{2}\vec{u}$ have the sense of $\vec{u}$; $-\vec{u}$ has the opposite sense.

Product of a vector by a real number

I is the midpoint of $[AB]$ if and only if $\overrightarrow{AI} = \overrightarrow{IB}$, or equivalently $\overrightarrow{AI} = \frac{1}{2}\overrightarrow{AB}$, or $\overrightarrow{IA} + \overrightarrow{IB} = \vec{0}$.

Product of a vector by a real number

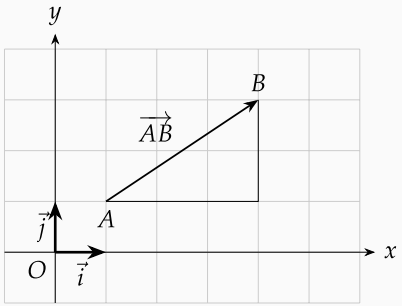
In a triangle ABC, I is the midpoint of $[AB]$ and J that of $[AC]$. Show that $\overrightarrow{IJ} = \frac{1}{2}\overrightarrow{BC}$.

Worked answer: Product of a vector by a real number

By Chasles, $\overrightarrow{IJ} = \overrightarrow{IA} + \overrightarrow{AJ} = -\frac{1}{2}\overrightarrow{AB} + \frac{1}{2}\overrightarrow{AC} = \frac{1}{2}(\overrightarrow{BA} + \overrightarrow{AC}) = \frac{1}{2}\overrightarrow{BC}$. We recover the parallelism of the lines (IJ) and (BC) when the triangle is not flat.

Frames, bases and coordinates

Two non-collinear vectors $\vec{i}, \vec{j}$ form a basis of the plane. Every vector $\vec{u}$ can be written uniquely as $\vec{u} = x\vec{i} + y\vec{j}$; its coordinates are (x, y) or $\begin{pmatrix} x \\ y \end{pmatrix}$. A frame $(O, \vec{i}, \vec{j})$ associates with the point M the coordinates of $\overrightarrow{OM}$. It is **orthonormal** if the basis vectors are perpendicular and of norm 1.



Frames, bases and coordinates

If $A(x_A, y_A)$, $B(x_B, y_B)$, $\vec{u}(x, y)$ and $\vec{v}(x', y')$:

$$\overrightarrow{AB} = \begin{pmatrix} x_B - x_A \\ y_B - y_A \end{pmatrix}, \quad \vec{u} + \vec{v} = \begin{pmatrix} x + x' \\ y + y' \end{pmatrix}, \quad k\vec{u} = \begin{pmatrix} kx \\ ky \end{pmatrix}.$$

Two vectors are equal if and only if their respective coordinates are equal.

For $\vec{u}(3,2)$ and $\vec{v}(-1,4)$, calculate $\vec{u}+\vec{v}$ and $2\vec{u}-\vec{v}$. Write $\vec{w}(5,-3)$ in the basis $(\vec{i},\vec{j})$.

Worked answer: Frames, bases and coordinates

$\vec{u} + \vec{v} = (2, 6)$; $2\vec{u} - \vec{v} = (7, 0)$; $\vec{w} = 5\vec{i} - 3\vec{j}$.

Norm, distance and midpoint

In an **orthonormal** frame, for $\vec{u}(x, y)$ and two points A, B:

$$\|\vec{u}\| = \sqrt{x^2 + y^2}, \qquad AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2}.$$

In any frame, the midpoint of $[AB]$ is $I\left(\frac{x_A + x_B}{2}; \frac{y_A + y_B}{2}\right)$.

Norm, distance and midpoint

The distance formula follows from Pythagoras's theorem applied to the horizontal and vertical displacements. For the midpoint, $\overrightarrow{AI} = \frac{1}{2}\overrightarrow{AB}$ gives $x_I - x_A = \frac{x_B - x_A}{2}$ and $y_I - y_A = \frac{y_B - y_A}{2}$; we isolate x_I, y_I.

Norm, distance and midpoint

In an orthonormal frame, $A(3,0)$, $B(5,3)$, $C(4,0)$ and $D(6,4)$.

1. Calculate $\overrightarrow{AB}$, $\overrightarrow{CD}$, then $3\overrightarrow{AB} - 5\overrightarrow{CD}$.
2. Determine U such that $\overrightarrow{BU} = 2\overrightarrow{AB} - \overrightarrow{CD}$.
3. Determine E such that $ABCE$ is a parallelogram, then its centre I.
4. Calculate AB and CD.

Worked answer: Norm, distance and midpoint

1. $\overrightarrow{AB} = (2, 3)$ and $\overrightarrow{CD} = (2, 4)$, so $3\overrightarrow{AB} - 5\overrightarrow{CD} = (-4, -11)$.
2. $\overrightarrow{BU} = (2, 2)$, so $U = B + (2, 2) = (7, 5)$.
3. $E = A + C - B = (2, -3)$. The centre is the midpoint of $[AC]$: $I(7/2, 0)$. The midpoint of $[BE]$ gives the same result.
4. $AB = \sqrt{13}$ and $CD = \sqrt{20} = 2\sqrt{5}$.

Collinearity and determinant

Two nonzero vectors are collinear if they have the same direction. The zero vector is collinear with every vector. If $\vec{v} \neq \vec{0}$, $\vec{u}$ and $\vec{v}$ are collinear if and only if there exists a real number k such that $\vec{u} = k\vec{v}$.

Collinearity and determinant

In the last characterisation, the condition $\vec{v} \neq \vec{0}$ is necessary: a nonzero vector cannot be a multiple of the zero vector.

Collinearity and determinant

For $\vec{u}(x, y)$, $\vec{v}(x', y')$, $\det(\vec{u}, \vec{v}) = xy' - yx'$.

$$\vec{u}, \vec{v} \text{ collinear} \iff \det(\vec{u}, \vec{v}) = 0.$$

Collinearity and determinant

If $\vec{u} = k\vec{v}$, then $xy' - yx' = kx'y' - ky'x' = 0$. The case where a vector is zero is immediate. Conversely, suppose $\vec{v} \neq \vec{0}$ and $xy' - yx' = 0$. If $x' \neq 0$, let $k = x/x'$. The equality gives $y = ky'$, so $\vec{u} = k\vec{v}$. If $x' = 0$, then $y' \neq 0$, hence $x = 0$; with $k = y/y'$, we again obtain $\vec{u} = k\vec{v}$.

Collinearity and determinant

Study the collinearity of: (a) $\vec{u}(3,2)$ and $\vec{v}(-6,-5)$; (b) $\vec{w}(4,1)$ and $\vec{t}(3,3/4)$. If $\overrightarrow{AB} = 2\overrightarrow{CD}$ and $\overrightarrow{BD} = -5\overrightarrow{CD}$, relate $\overrightarrow{AB}$ and $\overrightarrow{BD}$.

Worked answer: Collinearity and determinant

(a) $\det(\vec{u}, \vec{v}) = -15 + 12 = -3 \neq 0$: not collinear. (b) $\vec{t} = \frac{3}{4}\vec{w}$: collinear; the determinant is $4 \times \frac{3}{4} - 3 = 0$. Finally $\overrightarrow{AB} = -\frac{2}{5}\overrightarrow{BD}$, including if the vectors are zero.

Alignment and parallelism

For $A \neq B$ and $C \neq D$, the lines (AB) and (CD) are parallel (possibly coincident) if and only if $\det(\overrightarrow{AB}, \overrightarrow{CD}) = 0$. The points A, B, C are collinear if and only if $\det(\overrightarrow{AB}, \overrightarrow{AC}) = 0$.

Alignment and parallelism

Show that (AB) and (CD) are parallel with $A(2, -4)$, $B(10, -2)$, $C(-16, -4)$ and $D(12, 3)$.

Worked answer: Alignment and parallelism

$\overrightarrow{AB} = (8, 2)$ and $\overrightarrow{CD} = (28, 7)$. Their determinant is $8 \times 7 - 2 \times 28 = 0$. The points of each pair are distinct: both lines exist and are parallel.

Alignment and parallelism

Show that $E(0, 1/2)$, $F(2, 1)$ and $G(6, 2)$ are collinear.

Worked answer: Alignment and parallelism

$\overrightarrow{EF} = (2, 1/2)$ and $\overrightarrow{EG} = (6, 3/2) = 3\overrightarrow{EF}$. The determinant is $2 \times 3/2 - (1/2) \times 6 = 0$; so the points are collinear.

Alignment and parallelism

With $A(-1,1)$, $B(3,2)$, $C(-2,-3)$, $D(6,-1)$, $E(5,0)$, show that $(AB) \parallel (CD)$ and that E, B, D are collinear.

Worked answer: Alignment and parallelism

$\overrightarrow{AB} = (4, 1)$ and $\overrightarrow{CD} = (8, 2) = 2\overrightarrow{AB}$. Then $\overrightarrow{EB} = (-2, 2)$ and $\overrightarrow{ED} = (1, -1) = -\frac{1}{2}\overrightarrow{EB}$. Both results follow.